

Math 107 Midterm 1 practice

Last updated: September 24, 2026.

Based on [Lectures 01–14](#) and [Quizzes 1–4](#). Use with the [study guide](#). Exam: [Midterm 1](#) · Friday, October 2.

Three types of problems will appear on Midterm 1: short answer definitions and meaning, computational and applied problems.

A. Definitions and meaning

Write a short precise answer (one or two sentences, or a formula). No long essays.

- A1.** What is an n -vector? Give one non-math example from class (with units on the entries).
- A2.** What is a **linear combination** of vectors $a_1, \dots, a_k$?
- A3.** Define the **inner product** (dot product) of two n -vectors. When are two vectors **orthogonal**?
- A4.** When is a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ called **linear**? How does **affine** differ?
- A5.** Define the **Euclidean norm** $\|x\|$ and the **distance** between x and y .
- A6.** Write the formula for **cosine similarity** of nonzero a and b . What range of angles $\theta = \angle(a, b)$ do we use?
- A7.** What is a (**distance**) **nearest neighbor** of a query x among candidates $z_1, \dots, z_m$?
- A8.** In one sentence: what is the goal of **clustering** N vectors into k groups?
- A9.** State the definition of **linear dependence** of $a_1, \dots, a_k$. What does **independent** mean?
- A10.** Define $\text{span}\{a_1, \dots, a_k\}$.
- A11.** What is a **subspace** of $\mathbb{R}^n$? Why is the line $y = 2x + 1$ in $\mathbb{R}^2$ **not** a subspace?
- A12.** What is a **basis** of a subspace V ? What is $\dim V$?
- A13.** When is a list $q_1, \dots, q_k$ **orthonormal**? If it is an orthonormal basis of a subspace containing v , how do you find the coefficients of v ?
- A14.** True or false, with a one-line reason: “If $\|a - b\|$ is small, then $\angle(a, b)$ must be near 0.”
- A15.** True or false, with a one-line reason: “Every spanning list for V is a basis of V .”
- A16.** *k-means vocabulary.*
- (a) In k -means, what does the integer k stand for?
 - (b) After vectors have been assigned to k clusters, what is the formula for the **center (centroid)** of a nonempty cluster?
 - (c) Name the **two steps** that k -means alternates after an initial assignment (or after choosing initial centers).
- A17.** For each expression, say whether the result is a **scalar**, an n -**vector**, a **set of vectors**, or **undefined** (brief reason if undefined). Assume a, b, u, v are n -vectors and α is a scalar, unless noted.
- (a) $a^\top b$ (dot product of two n -vectors)
 - (b) $\|a\|$
 - (c) $a + b$
 - (d) αa
 - (e) cosine similarity of nonzero a and b
 - (f) $\angle(a, b)$
 - (g) $\text{span}\{a, b\}$
 - (h) $a^\top b + c$ when c is an n -vector

- (i) $a + b^\top$ (treating $b^\top$ as a row)
 - (j) $\text{rms}(a)$
 - (k) a k -means centroid of a nonempty cluster of n -vectors
 - (l) $\dim \text{span}\{a, b\}$
 - (m) $a^\top b$ when a is a 2-vector and b is a 3-vector
 - (n) unit vector $\hat{a} = a/\|a\|$ when $a \neq 0$
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B. Computational

Show your work for each calculation.

Vectors, combinations, dot products

- B1.** Let $u = (3, -1, 2)$ and $v = (0, 4, -1)$. Compute $u + v$, $2u - v$, and $u^\top v$.
- B2.** Write $w = (5, 1)$ as a linear combination of $a = (1, 0)$ and $b = (1, 1)$.
- B3.** Are $(1, 2)$ and $(-2, -4)$ orthogonal? Compute the dot product.
- B4.** Let $x = (2, 1, -1)$. Compute $\|x\|$ exactly (simplified radical OK).
- B5.** Distance between $p = (1, 1)$ and $q = (4, -3)$.
- B6.** Cosine of the angle between $a = (1, 0)$ and $b = (1, 1)$. Angle in degrees?
- B7.** Unit vector in the direction of $(3, 4)$.
- B8.** $x = (1, -1, 1, -1)$. Compute $\text{avg}(x)$ and $\text{rms}(x)$.
- B9.** Is $f(x) = 3x_1 - x_2 + 2$ linear or affine? Brief reason.
- B10.** Is $g(x) = x_1 x_2$ linear? Test with one superposition counterexample or explain.
- B11.** Score $s = w^\top r$ with $w = (0.5, 0.5)$ and $r = (80, 90)$. What is s ?
- B12.** Boolean vectors $x = (1, 0, 1, 1)$ and $y = (1, 1, 0, 1)$. What is $\|x - y\|$?

Nearest neighbors, scale, angle vs distance

- B13.** Query $x = (1, 2)$; candidates $z_1 = (0, 2)$, $z_2 = (3, 2)$, $z_3 = (1, 3)$. Which is nearest in Euclidean distance? Give the three distances.
- B14.** Same x and candidates as B13, but replace every first coordinate t by $10t$. Which is nearest now? Did it change?
- B15.** $x = (1, 0)$, $z_1 = (2, 0)$, $z_2 = (0.5, 0.5)$. Distance-NN and angle-NN of x ? Same or not?
- B16.** Cosine similarity of $(2, 1)$ and $(4, 2)$. What does that tell you about the angle?

***k*-means**

B17. Points $x_1 = (0, 0)$, $x_2 = (0, 2)$, $x_3 = (6, 0)$, $x_4 = (6, 2)$; representatives $z_1 = (1, 1)$, $z_2 = (5, 1)$. Assign each point to the nearest z_j .

B18. Using your clusters from B17, update each z_j to the mean of its points.

B19. After B18, do one more assignment step. Does any point switch clusters?

B20. A cluster of 2-vectors has points $(2, 0)$ and $(4, 2)$. What is the centroid?

B31. *k-means worksheet (2-vectors)*. Five vectors are initially assigned to $k = 2$ clusters as in the table. Compute the centroids, fill the distance table (Euclidean), then revise each cluster ID. Some entries are already filled — complete the blanks.

Vector	Initial Cluster ID	Revised Cluster ID
$x_1 = (0, 1)$	1	
$x_2 = (2, 1)$	1	
$x_3 = (8, 1)$	2	2
$x_4 = (9, 1)$	2	
$x_5 = (-2, 1)$	1	

Center of initial cluster 1: $c_1 =$ _____

Center of initial cluster 2: $c_2 = (8.5, 1)$ (given)

Distances	x_1	x_2	x_3	x_4	x_5
c_1		2			
c_2	8.5		0.5		10.5

(Exact values or one decimal place are fine.)

B32. *k-means worksheet (3-vectors)*. Four vectors in $\mathbb{R}^3$, $k = 2$. Compute both centroids from the initial assignment, fill **all** distances, then revise the cluster IDs.

Vector	Initial Cluster ID	Revised Cluster ID
$x_1 = (0, 0, 0)$	1	
$x_2 = (6, 0, 0)$	1	
$x_3 = (1, 0, 0)$	2	
$x_4 = (0, 8, 0)$	2	

$c_1 =$ _____

$c_2 =$ _____

Distances	x_1	x_2	x_3	x_4
c_1				
c_2				

Dependence, span, basis

- B21.** Are $(1, 0)$ and $(0, 1)$ independent in $\mathbb{R}^2$? Brief reason.
- B22.** Are $(2, 4)$ and $(1, 2)$ independent? If dependent, give a dependence relation.
- B23.** Find scalars not all zero with $\beta_1(1, 1) + \beta_2(1, -1) + \beta_3(0, 2) = 0$.
- B24.** Is $(3, 3)$ in $\text{span}\{(1, 1)\}$? If yes, give the coefficient.
- B25.** Is $(1, 0)$ in $\text{span}\{(1, 1), (1, -1)\}$? Coefficients or reason.
- B26.** Is $(1, 0, 0)$ in $\text{span}\{(1, 1, 0), (0, 1, 1)\}$? Reason.
- B27.** Give a basis for $\text{span}\{(1, 2, 0), (2, 4, 0), (0, 0, 1)\}$ and state the dimension.
- B28.** Is $\{(1, 0), (0, 1), (1, 1)\}$ a basis of $\mathbb{R}^2$? Why or why not?
- B29.** Are $q_1 = (1, 0)$ and $q_2 = (0, 1)$ orthonormal? For $v = (3, -2)$, find $\beta_1 = q_1^\top v$ and $\beta_2 = q_2^\top v$.
- B30.** Are $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$ orthonormal?
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C. Applied

Show reasoning. These mix ideas or need a careful choice of tool.

- C1.** Let $p = (1, 1)$, $q = (10, 10)$, $r = (1, -1)$.
- (a) Which of q, r is closer to p in **Euclidean distance**?
 - (b) Which is more similar to p in **direction**?
 - (c) In one sentence: when would you prefer angle over distance for “similarity”?
- C2.** A café stores each customer as (coffee count, pastry count). Someone adds a third feature “total items” = coffee + pastry.
- (a) Are the three feature directions dependent? Explain.
 - (b) Why might this hurt nearest-neighbor recommendations that use Euclidean distance?
- C3.** Data: $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (0, 1)$, $x_4 = (10, 10)$. Run one k -means **assignment** with $k = 2$ and initial $z_1 = (0, 0)$, $z_2 = (9, 9)$. Which points go to which cluster? What goes wrong conceptually if you wanted two “local” clusters near the origin and ignored the outlier’s pull on a centroid later?
- C4.** Let $a = (1, 2)$, $b = (2, 1)$, $c = (3, 3)$.
- (a) Show $\{a, b, c\}$ is dependent by exhibiting a dependence relation.
 - (b) Show $\{a, b\}$ is a basis of $\mathbb{R}^2$.
 - (c) Express c in that basis (coefficients).
- C5.** Decide which are subspaces of $\mathbb{R}^2$: (i) $\text{span}\{(2, 1)\}$; (ii) all (x, y) with $y = 3x + 1$; (iii) $\{(0, 0)\}$; (iv) the unit circle. One sentence each.
- C6.** Feature vectors for documents (word counts) can have very different total lengths. Why is **cosine similarity** often preferred to Euclidean distance for “similar documents”?
- C7.** $f(x) = \|x - c\|^2 - \|x - d\|^2$ for fixed c, d . Expand and show f is **affine** in x (find a and b in $a^\top x + b$).
- C8.** You normalize every candidate to unit length, then pick the Euclidean nearest neighbor to a query x that is already a unit vector. What comparison are you effectively making?
- C9.** Let $V = \text{span}\{(1, 0, 1), (0, 1, 1)\}$.
- (a) Find $\dim V$.
 - (b) Give one vector in $\mathbb{R}^3$ **not** in V , with a short reason.
 - (c) Is $(1, 1, 2)$ in V ? If yes, give coefficients.

C10. Mixed “which tool?” — for each task, name the best primary tool from: linear combination, dot product, norm/distance, angle, nearest neighbor, k -means, span, independence, basis.

Task	Tool
Weighted course score from a grade vector	
“Is this third sensor redundant?”	
“Which past day is most like today?”	
“Group songs into two mood types”	
“Can this vector be built from a and b ?”	
“Minimal independent spanning set for a plane through the origin”	

C11. *Graph — span.* In $\mathbb{R}^2$, sketch axes and draw $a = (2, 1)$ and $b = (-1, 2)$ as arrows from the origin.

- Sketch the line $\text{span}\{a\}$ (the whole line through a , not just the arrow).
- Shade or describe $\text{span}\{a, b\}$. Is it all of $\mathbb{R}^2$? Why?
- Mark a point v that is in $\text{span}\{a\}$ and a point w that is **not** in $\text{span}\{a\}$.

C12. *Graph — unit vectors and angle.* Let $u = (3, 0)$ and $v = (0, 4)$.

- Sketch u , v , and the unit vectors $\hat{u} = u/\|u\|$ and $\hat{v} = v/\|v\|$ from the origin (same axes).
- What are $\|\hat{u}\|$ and $\|\hat{v}\|$? What is $\angle(\hat{u}, \hat{v})$?
- On your sketch, which looks “closer” to $\hat{u}$: the tip of $\hat{v}$, or the tip of v ? Which comparison is Euclidean distance making? Which is angle making?

C13. *Graph — closeness vs direction.* Let $p = (1, 1)$, $q = (3, 3)$, and $r = (1, 0)$.

- (a) Sketch p , q , and r from the origin.
- (b) By eye (then check with a short computation): which of q, r is closer to p in **distance**? Which is closer in **angle/direction**?
- (c) Draw the unit circle lightly and mark $\hat{p}$, $\hat{q}$, $\hat{r}$. Explain in one sentence how the unit circle helps you see directional similarity.

Answers

Check only after a closed-book attempt. Update your definitions/formulas list for every miss.

A — Definitions (key points)

A1. List of n numbers (entries) with meaning/units; e.g. temperatures by hour, or (lat, lon).

A2. $\beta_1 a_1 + \dots + \beta_k a_k$ for scalars β_i .

A3. $a^\top b = \sum_i a_i b_i$; orthogonal iff $a^\top b = 0$.

A4. Linear: $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$ (equivalently $f(x) = a^\top x$). Affine: $a^\top x + b$ with possible nonzero b .

A5. $\|x\| = \sqrt{x^\top x}$; distance $\|x - y\|$.

A6. $a^\top b / (\|a\| \|b\|)$; $\theta \in [0, \pi]$.

A7. Any z_j minimizing $\|x - z_j\|$.

A8. Partition so same-group vectors are close (small within-group distances), with representatives.

A9. Dependent: nontrivial β with $\sum \beta_i a_i = 0$. Independent: only $\beta = 0$ works.

A10. All linear combinations of $a_1, \dots, a_k$.

A11. Nonempty set closed under linear combinations (contains 0). Line $y = 2x + 1$ misses 0 / not closed under scaling.

A12. Independent spanning list; dimension = number of vectors in any basis.

A13. Unit and pairwise orthogonal; $\beta_i = q_i^\top v$.

A14. **False** — e.g. near-origin opposite directions.

A15. False — spanning lists can be dependent.

A16. (a) The number of clusters (groups). (b) The mean of the vectors in that cluster: if cluster j contains $x_{i_1}, \dots, x_{i_m}$, then $c_j = \frac{1}{m}(x_{i_1} + \dots + x_{i_m})$. (c) **Assignment** (each point to the nearest current center) and **update** (replace each center by the mean of its cluster).

A17. (a) **scalar** (b) **scalar** (c) **n -vector** (d) **n -vector** (e) **scalar** (f) **scalar** (an angle, in radians or degrees) (g) **set of vectors** (a subspace / span) (h) **undefined** — scalar plus vector (i) **undefined** (sizes/shapes don't match for vector + row; not a valid n -vector sum in this course) (j) **scalar** (k) **n -vector** (l) **scalar** (a nonnegative integer) (m) **undefined** — lengths must match for a dot product (n) **n -vector**

B — Easy computations

B1. $(3, 3, 1)$; $(6, -6, 5)$; -6 .

B2. $w = 4a + 1b$ (or $w = 4(1, 0) + (1, 1)$).

B3. Dot $-2 - 8 = -10 \neq 0$; not orthogonal. (They are parallel / dependent.)

B4. $\sqrt{4+1+1} = \sqrt{6}$.

B5. $\|(-3, 4)\| = 5$.

B6. $1/\sqrt{2}$; 45° .

B7. $(3/5, 4/5)$.

B8. avg = 0; rms = 1.

B9. Affine (nonzero constant term).

B10. Not linear (product); e.g. $g(e_1) = 0 = g(e_2)$ but $g(e_1 + e_2) = 1$.

B11. 85.

B12. Disagree in 2 positions; $\|x - y\| = \sqrt{2}$.

B13. Distances 1, 2, 1; tie between z_1 and z_3 .

B14. After $\times 10$ on the first feature: distances 10, 20, 1; nearest is z_3 only. **Yes**, changed.

B15. Distance-NN z_2 ; angle-NN z_1 ; **not** the same.

B16. Cosine = 1; angle 0° (same direction).

B17. Cluster 1: $\{x_1, x_2\}$; cluster 2: $\{x_3, x_4\}$.

B18. $z_1 = (0, 1)$, $z_2 = (6, 1)$.

B19. Same assignment; no switches.

B20. $(3, 1)$.

B31. Initial cluster 1 is $\{x_1, x_2, x_5\}$, so

$$c_1 = \frac{(0, 1) + (2, 1) + (-2, 1)}{3} = (0, 1).$$

Distances:

	x_1	x_2	x_3	x_4	x_5
c_1	0	2	8	9	2
c_2	8.5	6.5	0.5	0.5	10.5

Revised IDs: 1, 1, 2, 2, 1 (no switches).

B32. $c_1 = \frac{1}{2}((0, 0, 0) + (6, 0, 0)) = (3, 0, 0);$
 $c_2 = \frac{1}{2}((1, 0, 0) + (0, 8, 0)) = (0.5, 4, 0).$

	x_1	x_2	x_3	x_4
c_1	3	3	2	$\sqrt{9+64} = \sqrt{73} \approx 8.5$
c_2	$\sqrt{0.25+16} = \sqrt{16.25} \approx 4.0$	$\sqrt{30.25+16} = \sqrt{46.25} \approx 6.8$	$\sqrt{0.25+16} \approx 4.0$	$\sqrt{0.25+16} \approx 4.0$

Revised: $x_1 \rightarrow 1, x_2 \rightarrow 1, x_3 \rightarrow 1$ (switches from 2), $x_4 \rightarrow 2$.

B21. Yes — standard basis; only the trivial combination gives 0.

B22. Dependent: e.g. $(2, 4) - 2(1, 2) = 0$.

B23. e.g. $\beta = (1, -1, 1)$ because $(1, 1) - (1, -1) = (0, 2)$.

B24. Yes: $3 \cdot (1, 1)$.

B25. Yes: $\frac{1}{2}(1, 1) + \frac{1}{2}(1, -1)$.

B26. No (inconsistent third coordinate).

B27. e.g. $\{(1, 2, 0), (0, 0, 1)\}$; $\dim = 2$.

B28. No — dependent (third = sum of first two), though it spans $\mathbb{R}^2$.

B29. Yes; $\beta_1 = 3, \beta_2 = -2$.

B30. Yes.

C — Harder

C1. (a) r . (b) q . (c) When overall scale should not matter.

C2. (a) Yes — third feature = first + second. (b) Double-counts information and can distort Euclidean distance.

C3. $x_1, x_2, x_3 \rightarrow z_1; x_4 \rightarrow z_2$. Clear discussion of outliers / initialization / choice of k is fine.

C4. (a) $a + b - c = 0$. (b) Two independent vectors in $\mathbb{R}^2$. (c) $c = a + b$.

C5. (i) yes; (ii) no; (iii) yes; (iv) no.

C6. Document lengths vary; cosine compares direction / word pattern.

C7. $f(x) = 2(d - c)^T x + \|c\|^2 - \|d\|^2$.

C8. Angle nearest-neighbor among those candidates.

C9. (a) 2. (b) e.g. $(1, -1, 0)$ or $(0, 0, 1)$. (c) Yes: $(1, 0, 1) + (0, 1, 1)$.

C10. Dot product; independence/dependence; nearest neighbor; k -means; span; basis.

C11. (a) Line through the origin along $(2, 1)$. (b) All of $\mathbb{R}^2$ — a, b are independent (not parallel). (c) e.g. $v = (4, 2)$ on the line; $w = (0, 1)$ off the line.

C12. (a) $\hat{u} = (1, 0), \hat{v} = (0, 1)$. (b) Both norms 1; angle 90° . (c) Tip of $\hat{v}$ is distance $\sqrt{2}$ from tip of $\hat{u}$; tip of v is farther from tip of $\hat{u}$. Distance compares positions of tips; angle compares directions from the origin (here both pairs of directions are orthogonal).

C13. (a) Sketch. (b) Distance: r is closer ($\|p - r\| = 1$ vs $\|p - q\| = 2\sqrt{2}$). Direction: q (same ray as p ; angle 0). (c) On the unit circle, $\hat{p} = \hat{q}$ coincide; $\hat{r}$ is elsewhere — directional closeness is closeness of the unit tips.